

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

A company has a newsletter. In January **2018**, there were **1,300** customers subscribed to the newsletter. For the next **24** months after January **2018**, the total number of customers subscribed to the newsletter each month was **7%** greater than the total number subscribed the previous month. Which equation gives the total number of customers, c , subscribed to the company's newsletter m months after January **2018**, where $m \leq 24$?

Answer

- A. $c = 1,300(0.07)^m$
- B. $c = 1,300(1.07)^m$
- C. $c = 1,300(1.7)^m$
- D. $c = 1,300(7)^m$

Correct Answer: B

Rationale

Choice B is correct. It's given that in January **2018**, there were **1,300** customers subscribed to a company's newsletter and for the next **24** months after January **2018**, the total number of customers subscribed to the newsletter each month was **7%** greater than the total number subscribed the previous month. It follows that this situation can be represented by the equation $c = a\left(1 + \frac{r}{100}\right)^m$, where c is the total number of customers subscribed to the company's newsletter m months after January **2018**, a is the number of customers subscribed to the newsletter in January **2018**, and the total number of customers subscribed to the newsletter each month was $r\%$ greater than the total number subscribed the previous month. Substituting **1,300** for a and **7** for r in this equation yields $c = 1,300\left(1 + \frac{7}{100}\right)^m$, or $c = 1,300(1.07)^m$.

Choice A is incorrect. This equation represents a situation where the total number of customers subscribed each month was **93%** less, not **7%** greater, than the total number subscribed the previous month.

Choice C is incorrect. This equation represents a situation where the total number of customers subscribed each month was **70%**, not **7%**, greater than the total number subscribed the previous month.

Choice D is incorrect. This equation represents a situation where the total number of customers subscribed each month was **600%**, not **7%**, greater than the total number subscribed the previous month.